

Individual Assessment Coversheet

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Group: 4

Lecturer's Name: Enos Nemukula

Student Full Name: Keenan Paiva Batista

Student Number: 4845350

Indicate	Yes	No
Plagiarism report attached		

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Section A

Question 1

1.1a)

$$f(h) = (h + 1)^2$$

$$f(h) = h^2 + 2h + 1$$

$$f(-h) = (-h)^2 - 2h + 1 \quad f(-h) = h^2 - 2h + 1$$

$$f(-h) = f(h) \text{ It does not have even symmetry}$$

$$f(-h) = h^2 - 2h + 1$$

$$f(-h) = -1(h^2 + 2h - 1) \quad f(-h) = -f(h) \text{ It does not have odd symmetry}$$

1.1b)

$$f(x) = \frac{x^2 + 3x}{(3x + 1)^2}$$

$$f(x) = \frac{x^2 + 3x}{9x^2 + 6x + 1}$$

$$f(-x) = \frac{(-x)^2 - 3x}{(-9x)^2 - 6x + 1}$$

$$f(-x) = \frac{x^2 - 3x}{9x^2 - 6x + 1}$$

$$f(-x) = f(x) \text{ It does not have even symmetry}$$

$$f(-x) = \frac{x^2 - 3x}{9x^2 - 6x + 1}$$

$$f(-x) = -1\left(\frac{x^2 + 3x}{9x^2 + 6x + 1}\right)$$

$$f(-x) = -f(x) \text{ It does not have odd symmetry (The Organic Chemistry Tutor, 2017)}$$

1.2a)

$$f(g(-4))$$

$$= f((-4)^2 - 1)$$

$$= f(16 - 1)$$

$$= f(15)$$

$$= \sqrt{15}$$

1.2b)

$$f(g(a))$$

$$= f(a^2 - 1)$$

$$= \sqrt{a^2 - 1}$$

1.2c)

$$g(f(a))$$

$$= g(\sqrt{a})$$

$$= (\sqrt{a})^2 - 1$$

$$= a - 1$$

Question 2

2.1)

In terms of a function, the domain represents the possible input values, the number of users, while the range represents the possible output values, the response time of the web server. This indicates that the response time of the web server is completely dependent on the number of users accessing the web server at the same time. Depending on the type of function used, a change to the number of users can increase or decrease the response time of the server.

2.2)

$$f(x) = \frac{x^3 + 3x^2 + 6}{x^2 + x - 1}$$

$$\begin{aligned} x^2 + x - 1 &\neq 0 \\ \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ \frac{-1 \pm \sqrt{1^2 - 4(1)(-1)}}{2(1)} \\ \frac{-1 \pm \sqrt{5}}{2} \end{aligned}$$

$$x = 0, 62 \text{ or } x = -1, 62$$

$$x \in \mathbb{R}; x \neq 0, 62 \text{ or } x \neq -1, 62$$

Question 3

3a)

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{(2x^2 + 1)^2}{(x - 1)^2(x^2 + x)} \\ &= \lim_{x \rightarrow \infty} \frac{((2x^2 + 1)^2)((2x^2 + 1)^2)}{(x - 1)(x - 1)(x^2 + x)} \\ &= \lim_{x \rightarrow \infty} \frac{4x^4 + 4x^2 + 1}{(x^2 - 2x + 1)(x^2 + x)} \\ &= \lim_{x \rightarrow \infty} \frac{4x^4 + 4x^2 + 1}{x^4 + x^3 - 2x^3 + 2x^2 + x^2 + x} \\ &= \lim_{x \rightarrow \infty} \frac{4x^4 + 4x^2 + 1}{x^4 - x^3 + 3x^2 + x} \\ &= \lim_{x \rightarrow \infty} \frac{\frac{4x^4}{x^4} + \frac{4x^2}{x^4} + \frac{1}{x^4}}{\frac{x^4}{x^4} - \frac{x^3}{x^4} + \frac{3x^2}{x^4} + \frac{x}{x^4}} \\ &= \lim_{x \rightarrow \infty} \frac{4 + \frac{4}{x^2} + \frac{1}{x^4}}{1 - \frac{1}{x} + \frac{3x}{x^2} + \frac{1}{x^3}} \\ &= \frac{4 + 0 + 0}{1 - 0 + 0 + 0} \\ &= 4 \end{aligned}$$

3b)

$$\begin{aligned}
 & \lim_{x \rightarrow 4} \frac{4-x}{5-\sqrt{x^2+9}} \\
 &= \lim_{x \rightarrow 4} \frac{4-x}{5-\sqrt{x^2+9}} \times \frac{5+\sqrt{x^2+9}}{5+\sqrt{x^2+9}} \\
 &= \lim_{x \rightarrow 4} \frac{(4-x)(5+\sqrt{x^2+9})}{(5-\sqrt{x^2+9})(5+\sqrt{x^2+9})} \\
 &= \lim_{x \rightarrow 4} \frac{(4-x)(5+\sqrt{x^2+9})}{25+5\sqrt{x^2+9}-5\sqrt{x^2+9}-x^2-9} \\
 &= \lim_{x \rightarrow 4} \frac{(4-x)(5+\sqrt{x^2+9})}{16-x^2} \\
 &= \lim_{x \rightarrow 4} \frac{(4-x)(5+\sqrt{x^2+9})}{(4-x)(4+x)} \\
 &= \lim_{x \rightarrow 4} \frac{5+\sqrt{x^2+9}}{4+x} \\
 &= \frac{5+\sqrt{4^2+9}}{4+4} \\
 &= \frac{10}{8} \\
 &= \frac{5}{4} \text{ (The Organic Chemistry Tutor, 2018)}
 \end{aligned}$$

3c)

$$\begin{aligned}
 & \lim_{x \rightarrow 0} \frac{x}{\sqrt{kx+1}-1} \\
 &= \lim_{x \rightarrow 0} \frac{x}{\sqrt{kx+1}-1} \times \frac{\sqrt{kx+1}+1}{\sqrt{kx+1}+1} \\
 &= \lim_{x \rightarrow 0} \frac{(x)(\sqrt{kx+1}+1)}{(\sqrt{kx+1}-1)(\sqrt{kx+1}+1)} \\
 &= \lim_{x \rightarrow 0} \frac{(x)(\sqrt{kx+1}+1)}{kx+1+\sqrt{kx+1}-\sqrt{kx+1}-1} \\
 &= \lim_{x \rightarrow 0} \frac{(x)(\sqrt{kx+1}+1)}{kx} \\
 &= \lim_{x \rightarrow 0} \frac{\sqrt{kx+1}+1}{k} \\
 &= \frac{\sqrt{k(0)+1}+1}{k} \\
 &= \frac{\sqrt{1+1}}{k} \\
 &= \frac{2}{k}
 \end{aligned}$$

3d)

$$\begin{aligned}
 & \lim_{a \rightarrow -\infty} \frac{-a^4-2a^2+1}{a-3a^4+5} \\
 &= \lim_{a \rightarrow -\infty} \frac{\frac{-a^4}{a^4}-\frac{2a^2}{a^4}+\frac{1}{a^4}}{\frac{a}{a^4}-\frac{3a^4}{a^4}+\frac{5}{a^4}} \\
 &= \lim_{a \rightarrow -\infty} \frac{-1-\frac{2}{a^2}+\frac{1}{a^4}}{\frac{1}{a^3}-3+\frac{5}{a^4}} \\
 &= \frac{-1+0-0}{-0-3-0} \\
 &= \frac{1}{3}
 \end{aligned}$$

Question 4

4.1a) $f(x) = \frac{1}{2\sqrt{3x+1}}$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{2\sqrt{3x+h+1}} - \frac{1}{2\sqrt{3x+1}}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{2\sqrt{3(0)+h+1}} - \frac{1}{2\sqrt{3(0)+1}}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{2\sqrt{h+1}} - \frac{1}{2}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{2\sqrt{h+1}} - \frac{1}{2}}{h} \times \frac{\frac{1}{2\sqrt{h+1}} + \frac{1}{2}}{\frac{1}{2\sqrt{h+1}} + \frac{1}{2}}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{4(h+1)} + \frac{1}{2} \left(\frac{1}{2\sqrt{h+1}} \right) - \frac{1}{2} \left(\frac{1}{2\sqrt{h+1}} \right) - \frac{1}{4}}{\frac{h}{2\sqrt{h+1}} + \frac{h}{2}}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{4h+4} - \frac{1}{4}}{\frac{h}{2\sqrt{h+1}} + \frac{h}{2}}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1-4h}{4h+4}}{\frac{h}{2\sqrt{h+1}} + \frac{h}{2}}$$

$$= \lim_{h \rightarrow 0} \frac{h \left(\frac{1-4}{4h+4} \right)}{h \left(\frac{1}{2\sqrt{h+1}} + \frac{1}{2} \right)}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{-3}{4h+4}}{\frac{1}{2\sqrt{h+1}} + \frac{1}{2}}$$

$$= \frac{\frac{-3}{4(0)+4}}{\frac{1}{2\sqrt{0+1}} + \frac{1}{2}}$$

$$= \frac{\frac{-3}{4}}{\frac{1}{2} + \frac{1}{2}}$$

$$= -\frac{3}{4}$$

$m = -\frac{3}{4}$ (toppr, 2024)

4.1b)

$$y = mx + c$$

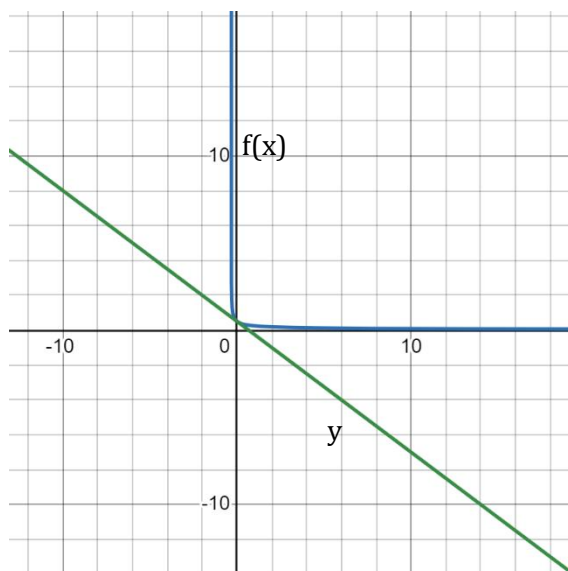
$$\frac{1}{2} = -\frac{3}{4}(0) + c$$

$$c = \frac{1}{2}$$

$$y = -\frac{3}{4}x + \frac{1}{2}$$

4.1c)

Made in desmos



4.1d)

The tangent of a function intersects with it at a point. We can use the tangent to plot the curve of the function in a graph. Additionally, the results that were obtained determines that the value of x must be more than -1 in order for function f(x) to be defined. We can also determine the equation of function f(x) makes it a rational function.

4.2)

$$y = \sqrt{4x} = (4x)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2}(4x)^{-\frac{1}{2}} \times 4$$

$$4 = \frac{1}{\sqrt{x}}$$

$$4\sqrt{x} = 1\sqrt{x} = \frac{1}{4}$$

$$x = \frac{1}{16}$$

$$n = \frac{1}{16}$$

$$Q(n; 2\sqrt{n})$$

$$Q(\frac{1}{16}; 2\sqrt{\frac{1}{16}})$$

$$Q(\frac{1}{16}; \frac{1}{2})$$

Question 5

5a)

$$f(x) = e^{2x}(2x - 7)^5$$

$$f'(x) = (e^{2x} \times 2)(2x - 7)^5 + (e^{2x})(5(2x - 7)^4 \times 2)$$

$$f'(x) = 2e^{2x}(2x - 7)^5 + 10e^{2x}(2x - 7)^4$$

$$f'(x) = 2e^{2x}((2x - 7)^5 + 5(2x - 7)^4)$$

5b)

$$f(x) = \frac{-2\ln(x)}{3x-1}$$

$$f'(x) = \frac{(-2 \times \frac{1}{x})(3x-1) - (-2\ln(x))(3)}{(3x-1)^2}$$

$$f'(x) = \frac{-6+2x^{-1}+6\ln(x)}{9x^2-6x+1}$$

$$f'(x) = \frac{6\ln(x) + \frac{2}{x} - 6}{9x^2 - 6x + 1}$$

(The Organic Chemistry Tutor, 2018)

5c)

$$k = 2\theta\sqrt{\cos\theta} = 2\theta(\cos\theta)^{\frac{1}{2}}$$

$$\frac{dk}{dx} = (2)(\cos\theta)^{\frac{1}{2}} + (2\theta)\left(\frac{1}{2}(\cos\theta)^{-\frac{1}{2}} \times -\sin\theta\right)$$

$$\frac{dk}{dx} = 2\sqrt{\cos\theta} + 2\theta\left(\frac{\frac{1}{2}}{\sqrt{\cos\theta}} \times -\frac{\sin\theta}{1}\right)$$

$$\frac{dk}{dx} = 2\sqrt{\cos\theta} + 2\theta\left(\frac{-\frac{1}{2}\sin\theta}{\sqrt{\cos\theta}}\right)$$

$$\frac{dk}{dx} = \frac{2\sqrt{\cos\theta}}{1} + \frac{-\theta\sin\theta}{\sqrt{\cos\theta}}$$

$$\frac{dk}{dx} = \frac{(2\sqrt{\cos\theta})(\sqrt{\cos\theta}) - \theta\sin\theta}{\sqrt{\cos\theta}}$$

$$\frac{dk}{dx} = \frac{2\cos\theta - \theta\sin\theta}{\sqrt{\cos\theta}}$$

Question 6

6a)

$f(x) = x + 2\cos x[-\pi; \pi]$ To find the critical points, we must find the first derivative.

$f'(x) = 1 - 2\sin x$ At the critical points $f'(x) = 0$

$$-1 = -2\sin x$$

$$\sin x = \frac{1}{2}$$

$$x = \frac{1}{6} \times \pi$$

$$x = \frac{\pi}{6} \text{ rad}$$

$$f(\pi) = \pi + 2\cos\pi = 1, 14$$

$$f(-\pi) = -\pi + 2\cos -\pi = -5, 14$$

$$f\left(\frac{\pi}{6}\right) = \frac{\pi}{6} + 2\cos\frac{\pi}{6} = 2, 26$$

$$\text{Maximum} = \left(\frac{\pi}{6}; 2, 26\right)$$

$$\text{Minimum} = (-\pi; -5, 14)$$

6b)

User experience is the most improved when the response time is at its peak, the data scientist can optimise the web application to keep its max response time of $\frac{\pi}{6}$; by keeping the function at 2,26.

Question 7

7a)

$$f(x) = \frac{x+8}{(3x+1)^2} (0, 8)$$

$$f(0) = \frac{0+8}{(3(0)+1)^2} = \frac{8}{1} = 8$$

$$f(8) = \frac{8+8}{(3(8)+1)^2} = \frac{16}{625}$$

$$\frac{f(x)-f(a)}{x-a}$$

$$= \frac{f(8)-f(0)}{8-0}$$

$$= \frac{\frac{16}{625}-8}{8}$$

$$= -\frac{123}{125}$$

The rate of change is $-\frac{123}{125} m.s^{-2}$ (The Organic Chemistry Tutor, 2016)

7b)

The slope of a function is used to show the steepness and direction of the function in a graph. Therefore, the slope can't be used in an equation for vertical lines. This is due to vertical lines not having slopes.

Bibliography

The Organic Chemistry Tutor, 2016. *Average and Instantaneous Rate of Change of a function over an interval & a point - Calculus*. [Online]

Available at: <https://www.youtube.com/watch?v=4Up5gsDeluw>

[Accessed 21 June 2024].

The Organic Chemistry Tutor, 2017. *Even, Odd, or Neither Functions The Easy Way! - Graphs & Algebraically, Properties & Symmetry*. [Online]

Available at: <https://www.youtube.com/watch?v=fKyBOLsqRlo>

[Accessed 18 June 2024].

The Organic Chemistry Tutor, 2018. *Derivative of Logarithmic Functions*. [Online]

Available at: <https://www.youtube.com/watch?v=Dp9sgIvaKPk>

[Accessed 20 June 2024].

The Organic Chemistry Tutor, 2018. *Evaluating Limits With Fractions and Square Roots*. [Online]

Available at: <https://www.youtube.com/@TheOrganicChemistryTutor>

[Accessed 19 June 2024].

toppr, 2024. *Derivatives*. [Online]

Available at: [https://www.toppr.com/guides/maths/limits-and-](https://www.toppr.com/guides/maths/limits-and-derivatives/derivatives/#:~:text=Answer%3A%20The%20derivative%20is%20what,that%20derivative%20is%20a%20slope.)

[derivatives/derivatives/#:~:text=Answer%3A%20The%20derivative%20is%20what,that%20derivative%20is%20a%20slope.](https://www.toppr.com/guides/maths/limits-and-derivatives/derivatives/#:~:text=Answer%3A%20The%20derivative%20is%20what,that%20derivative%20is%20a%20slope.)

[Accessed 19 June 2024].